Pinhole Sunglasses — Fresh Sweep Analysis

Blur-Optimization Model and Acuity Model, Recomputed From Scratch

All data in this document was generated in a single fresh run against the corrected optics (`stage2_fixed/`, post the lens-wavelength and import-shadowing fixes) — no figure or table reuses a previously-computed CSV.

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Chapter 1
Purpose and methodology

This document presents two theoretical models, swept fresh from scratch on the same 15×8 grid (15 apertures $\times$ 8 myopia levels), against this project's corrected optics pipeline:

- **The blur-optimization model** (`stage2_fixed/phase2_retina.py`): the exact scalar Angular Spectrum Method propagation from pinhole through the eye to the retina, reporting encircled-energy blur radius $r_{EE}(50\%)$ and the second-moment (RMS) spot radius.
- **The acuity model** (`stage4_human_vision/pipeline.py`): the same optical chain, extended through pupil/illuminance, contrast sensitivity, and Visual-Strehl-adjusted acuity prediction, reporting predicted logMAR.

Grid: aperture $a_{hole} \in \{75, 100, 150, 200, 300, 400, 500, 600, 700, 750, 1000, 1250, 1500, 1750, 2000\}$ μm (this project's standard list) $\times$ myopia $M \in \{0, 1, 2, 3, 4, 5, 6, 8\}$ D — 120 points per model, 240 total. Fixed parameters: wavelength 550 nm, pinhole-to-cornea gap 12 mm, natural pupil 4 mm, eye power $P_0 = 60$ D, grid 2048×2048 over a 6 mm window. **Both sweeps completed with zero junction-check failures and zero exceptions.**

On the corrected optics: this run is downstream of two fixes documented in `stage2_fixed/PHYSICS_AUDIT.md` and the `RESULTS.md` files it references: (1) a lens-phase/retina-propagation wavelength mismatch that underapplied the eye lens's focusing power by a factor of $n' \approx 1.336$ whenever `refractive_index_correction=True` (the default), and (2) a Python import-order bug that had silently redirected `stage4_human_vision`'s own optics import to a deprecated, still-buggy copy of the pipeline. Every number in this document reflects both fixes.

A third, separate acuity architecture (`R7acuity`, an ideal-observer Monte Carlo model) was also reattempted from scratch; its calibration step crashes against the corrected optics rather than producing a sweep — covered in Chapter 5, not silently omitted.

Chapter 2

The blur-optimization model

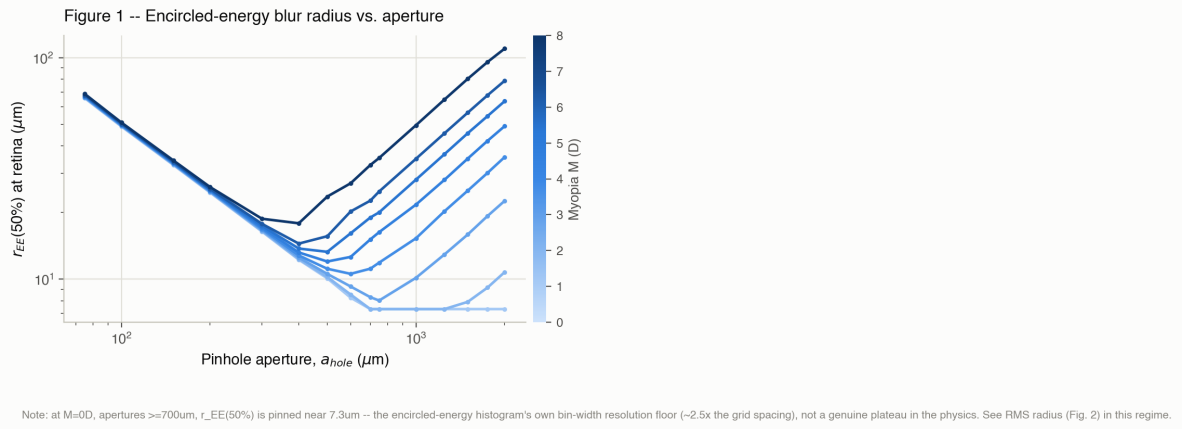


Figure 2.1: Encircled-energy blur radius $r_{EE}(50\%)$ at the retina, vs. aperture, one curve per myopia level (colorbar). Every curve is V-shaped: diffraction dominates on the left (blur falls as aperture grows), defocus dominates on the right (blur grows with aperture) once the aperture is admitting enough light to reveal the eye's own uncorrected myopic defocus.

A numerical caveat, not a physics finding: at $M = 0$D, for every aperture $\geq 700\,\mu$m, $r_{EE}(50\%)$ reads a constant $7.33\,\mu$m (visible as the flat tail on the lightest curve in Figure 1). This is the encircled-energy histogram's own bin width ($\approx 2.5\times$ the grid spacing $dx \approx 2.93\,\mu$m $\approx 7.3\,\mu$m) becoming the resolution floor once the true blur spot is this tight (large aperture, near-perfect focus) — confirmed by RMS radius (Figure 2), which keeps varying smoothly and continuously across the identical range. **RMS radius, not $r_{EE}(50\%)$, is the trustworthy metric wherever this floor is hit.** The same histogram-floor mechanism can in principle affect $r_{EE}(70/80/90\%)$ too, but at a larger radius (more energy to enclose needs more bins), so it binds less often at these percentiles — visible directly in Table 2.1 below (only the M=0–1D, $r_{EE}(50\%)$ cells hit exactly $7.33\,\mu$m; no other percentile/M combination does).

2.1 Optimum aperture by myopia level (pure blur minimization, all four percentiles)

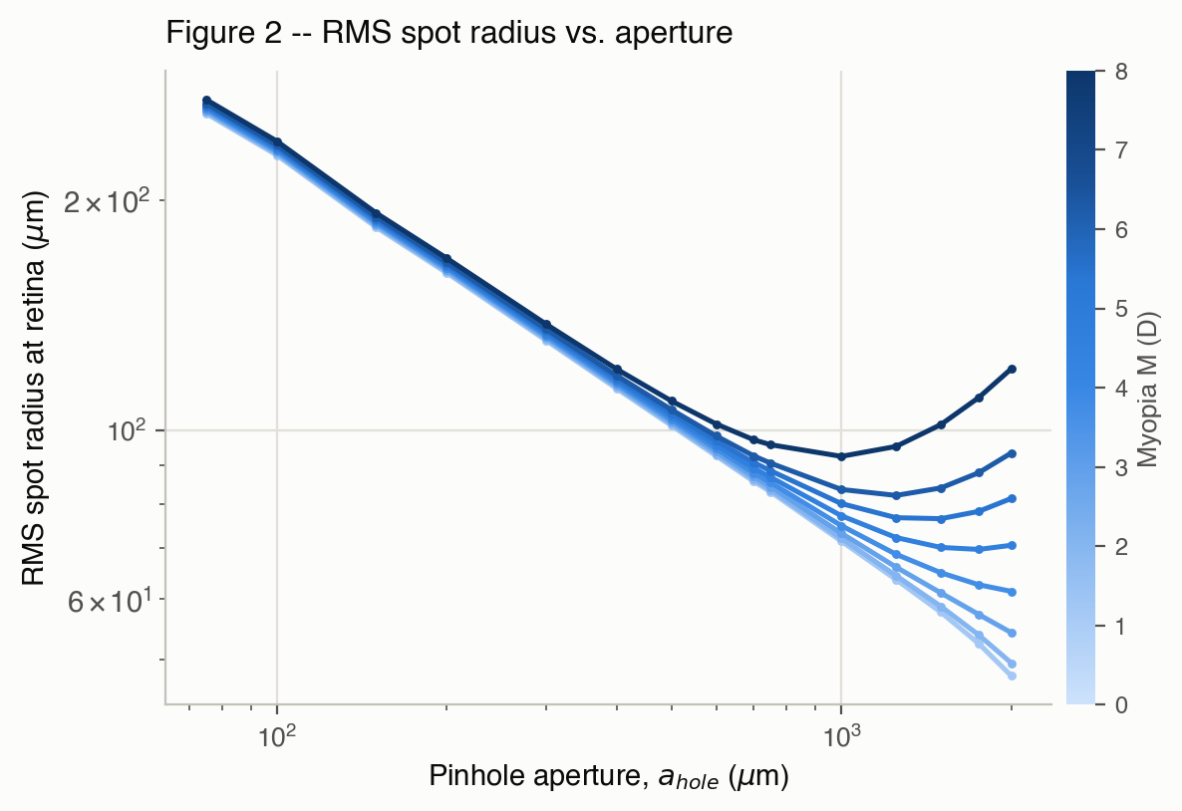


Figure 2.2: RMS spot radius at the retina, same grid. Unlike $r_{EE}(50\%)$, RMS radius has *not yet reached its minimum* by 2000 μm for $M \leq 6$D — it is still falling at the largest aperture tested. Only $M = 8$D shows a clear minimum within this range (∼1000–1250 μm). This is consistent with this project's own earlier finding (`model_comparison/README.md`) that RMS radius weights the diffuse outer Fresnel-ring energy more heavily than $r_{EE}(50\%)$ does, and tends to prefer larger apertures.

	$r_{EE}(50\%)$		$r_{EE}(70\%)$		$r_{EE}(80\%)$		$r_{EE}(90\%)$	
M (D)	opt. (μm)	val. (μm)	opt. (μm)	val. (μm)	opt. (μm)	val. (μm)	opt. (μm)	val. (μm)
0	700	7.33*	1000	7.33*	1250	7.33*	2000	10.50
1	700	7.33*	1000	9.24	1250	13.36	1250	20.15
2	750	8.01	600	13.49	750	19.08	1000	28.39
3	600	10.57	500	16.00	750	22.76	750	34.97
4	500	12.02	500	19.52	600	27.06	750	40.94
5	500	13.29	400	20.82	600	29.94	600	45.49
6	400	14.47	400	23.11	500	33.72	600	50.93
8	400	17.87	300	27.36	400	40.21	500	58.81

Table 2.1: Table 2.1 — Blur-only optimal aperture (the a_{hole} minimizing that column's percentile) and the blur radius achieved there, for all four encircled-energy percentiles, at every myopia level. *Sits inside the $r_{EE}(50\%)$ histogram-floor region (see caveat above) — lower-confidence. Full data: `table1_optimum_summary.csv`. Note the optimal aperture is not a single number even for pure blur minimization: it depends on which percentile of the light distribution is being minimized, growing substantially from $r_{EE}(50\%)$ to $r_{EE}(90\%)$ at every M tested (e.g. M=5D: 500 vs. 600 μm; M=0D: 700 vs. 2000 μm).

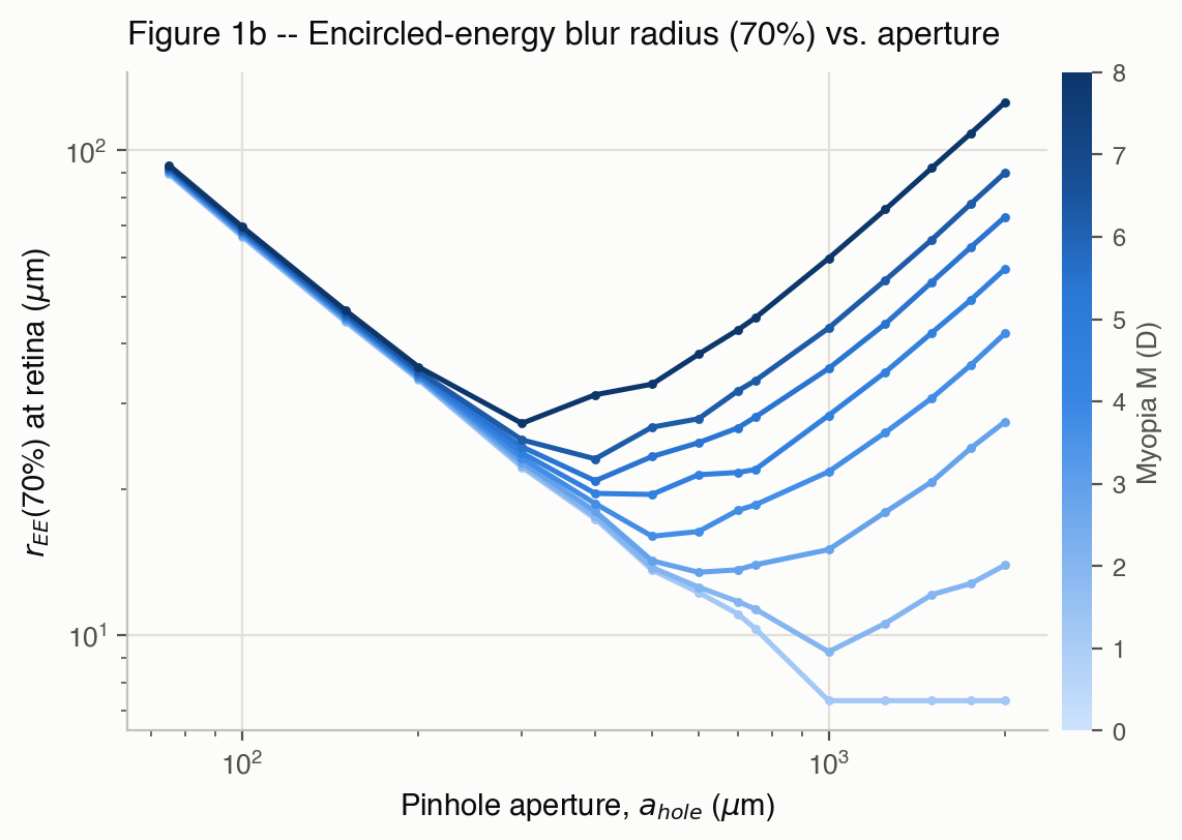


Figure 2.3: Encircled-energy blur radius $r_{EE}(70\%)$ vs. aperture, same grid.

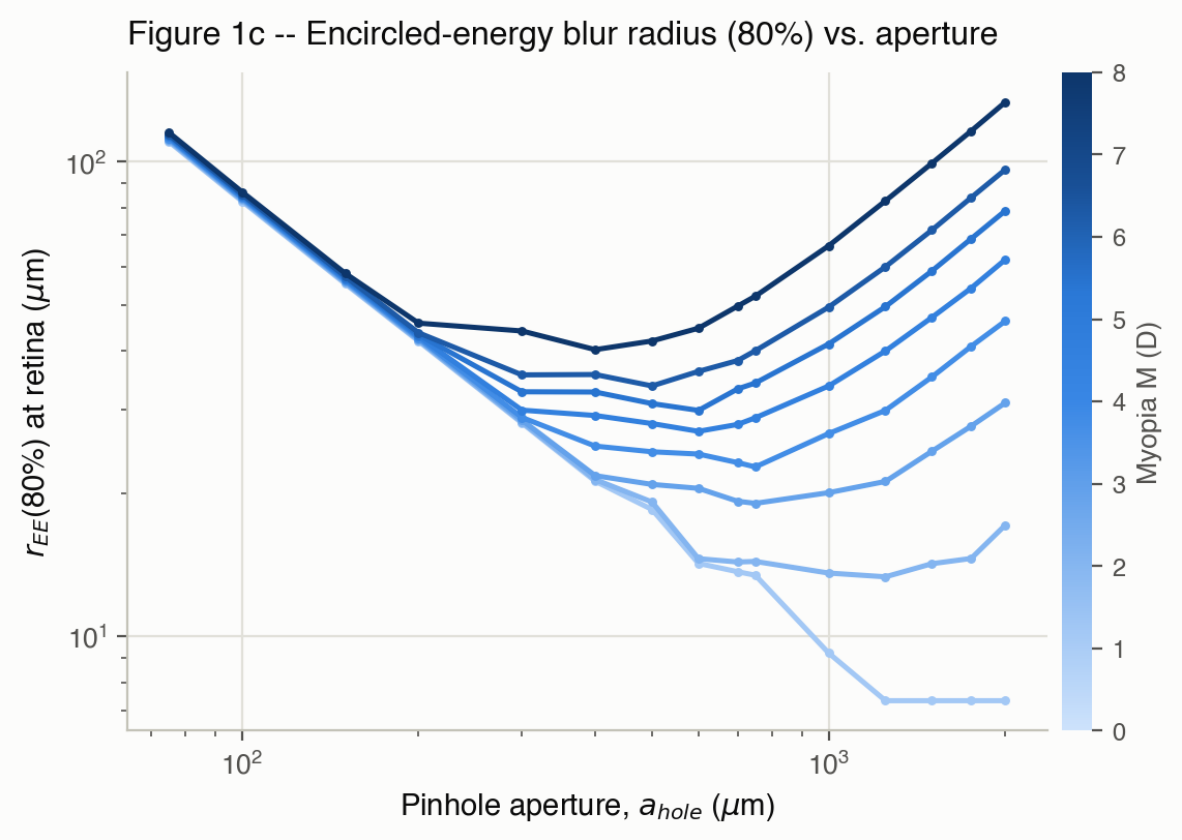


Figure 2.4: Encircled-energy blur radius $r_{EE}(80\%)$ vs. aperture, same grid.

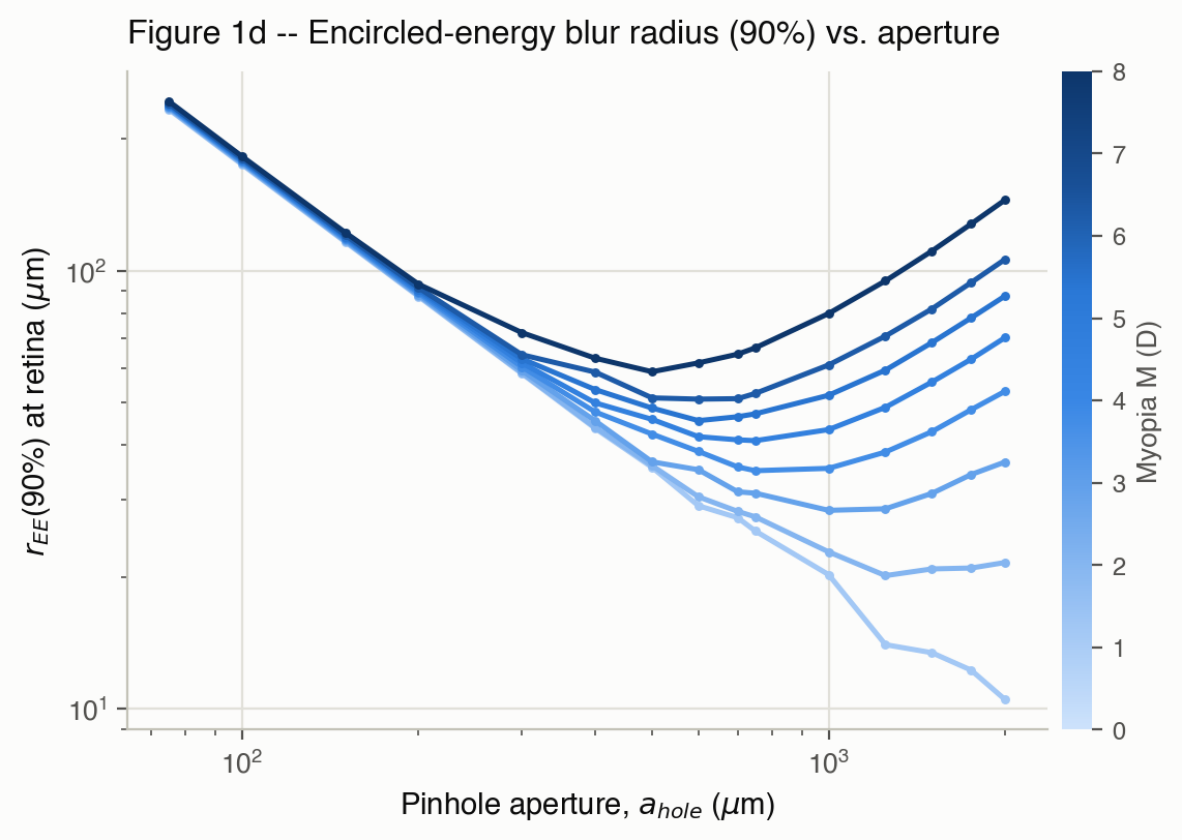


Figure 2.5: Encircled-energy blur radius $r_{EE}(90\%)$ vs. aperture, same grid. As the target percentile rises, the V-shaped minimum broadens and shifts to progressively larger apertures — capturing more of the diffuse outer Fresnel-ring energy increasingly rewards a larger, less severely diffracting aperture, the same effect RMS radius (Figure 2) shows in its own, stronger form.

Chapter 3
The acuity model

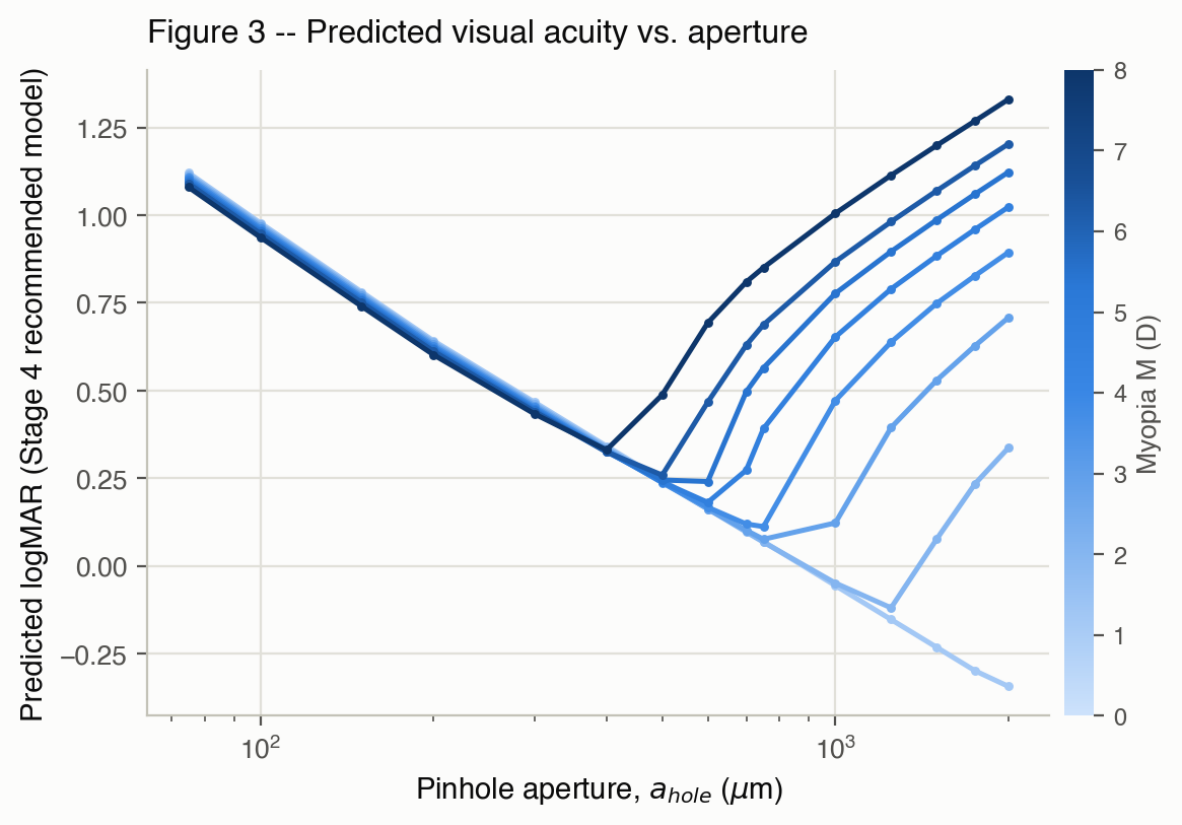


Figure 3.1: Stage 4's recommended predicted logMAR (MTF10/VSOTF/brightness-adjusted, lower is better) vs. aperture. The same qualitative V-shape as the blur curves, but the minima are visibly shifted toward *larger* apertures and are broader/shallower — the acuity model additionally accounts for retinal illuminance and contrast sensitivity, which favor more light throughput than pure blur minimization does.

3.1 Representative apertures at $M = 0, 2, 5$D

Recall from `stage4_human_vision/RESULTS.md` that this predictor, even against the corrected optics, does not achieve statistically significant agreement with the one available real clinical dataset (Wang 2009) at any myopia level tested — these logMAR values are internally-consistent theoretical predictions, not validated clinical forecasts. They are presented here as the model's own output, not as ground truth.

M (D)	a_{hole} (μm)	$r_{EE}(50\%)$ (μm)	RMS (μm)	logMAR	Snellen (20/x)
0	500	10.09	101.36	0.245	35.1
0	750	7.33*	82.92	0.069	23.4
0	1000	7.33*	71.48	−0.056	17.6
0	1250	7.33*	63.65	−0.153	14.1
0	1500	7.33*	57.70	−0.232	11.7
0	2000	7.33*	47.59	−0.344	9.1
2	500	10.56	102.51	0.238	34.6
2	750	8.01	84.22	0.076	23.8
2	1000	10.14	73.20	0.122	26.5
2	1250	12.93	66.08	0.394	49.6
2	1500	15.92	61.14	0.529	67.6
2	2000	22.55	54.22	0.708	102.1
5	500	13.29	105.16	0.245	35.1
5	750	20.03	88.38	0.565	73.4
5	1000	28.13	80.06	0.777	119.6
5	1250	36.66	76.73	0.896	157.5
5	1500	45.60	76.50	0.986	193.8
5	2000	63.54	81.32	1.123	265.2

Table 3.1: *Inside the $r_{EE}(50\%)$ histogram-floor region (Chapter 2); RMS radius is the reliable blur column for these rows.

Chapter 4
Comparing the two models

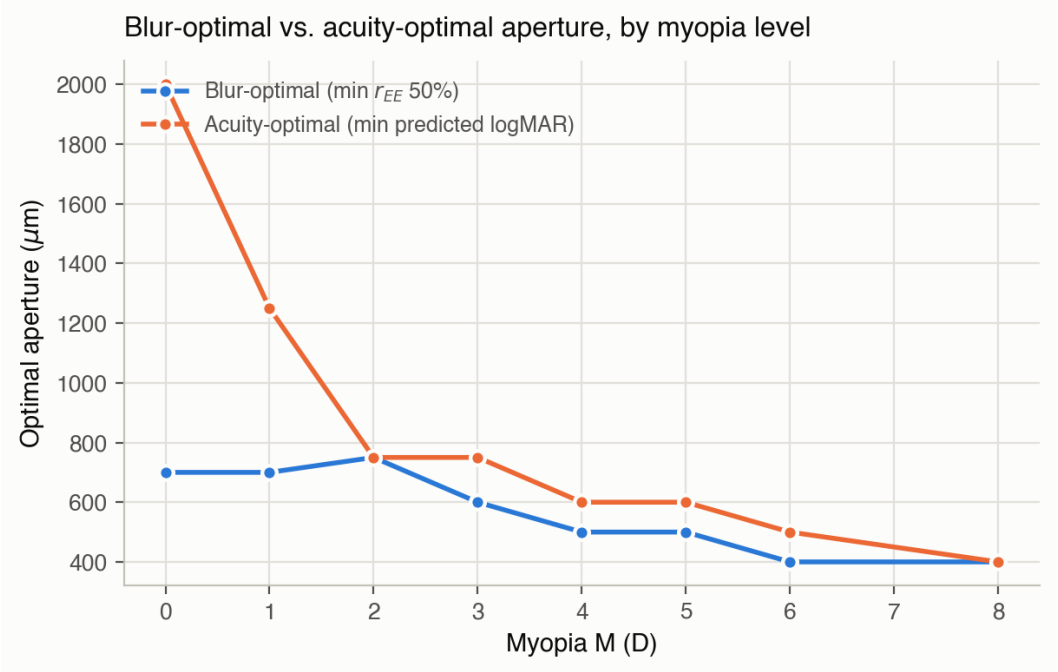


Figure 4.1: The blur-minimizing aperture and the acuity-minimizing aperture, both as a function of myopia. They diverge sharply at low myopia ($M \leq$ 1D: 700 μm vs. 1250–2000 μm) and converge as myopia increases, meeting exactly at $M = 8$D (400 μm both).

M (D)	Blur-opt. (μm)	$r_{EE}(50\%)$ there	Acuity-opt. (μm)	logMAR there
0	700	7.33*	2000	−0.344
1	700	7.33*	1250	−0.120
2	750	8.01	750	0.076
3	600	10.57	750	0.112
4	500	12.02	600	0.181
5	500	13.29	600	0.240
6	400	14.47	500	0.259
8	400	17.87	400	0.331

Table 4.1: *Histogram-floor region, see Chapter 2.

Why they diverge at low myopia. At M = 0–1D, defocus blur is small at almost every tested aperture (the eye is near-emmetropic), so pure blur minimization keeps favoring smaller apertures for their tighter diffraction spot. The acuity model additionally rewards the extra light throughput and contrast a larger aperture admits (via retinal illuminance and the CSF-weighted Visual Strehl term) once blur is no longer the binding constraint — so it keeps preferring larger apertures long after blur-minimization has stopped caring. **Why they converge at high myopia.** Once M is large, defocus blur dominates at every aperture in this range and grows sharply past each curve's minimum (Chapter 2, Figure 1) — both models

are now dominated by the same effect (uncorrected defocus) and agree on a small aperture.

Chapter 5

R7acuity: attempted, and why it produced no sweep

`R7acuity`'s ideal-observer Monte Carlo model was re-run from scratch against the corrected optics before any sweep was attempted. Its mandatory first step, calibrating a noise parameter against a fixed reference condition (3 mm aperture, $M = 0$, diffraction-limited, target 1.0 arcmin = 20/20), fails outright:

```
bracket search: sigma=0.000100 -> threshold=0.0556 arcmin
bracket search: sigma=1.000000 -> threshold=0.0556 arcmin
CALIBRATION FAILED (fresh run, from scratch): target 1.0 not bracketed by
[0.05559412519746301, 0.05559412519746301] at sigma in [0.0001, 1.0]
-- widen the search bracket
```

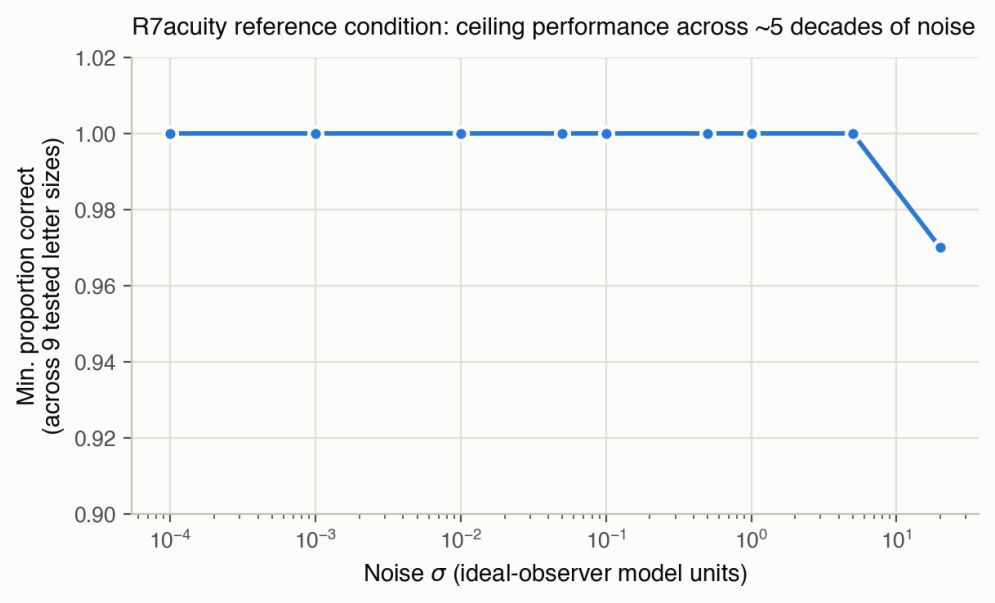


Figure 5.1: Direct probe of the reference condition: minimum proportion-correct across 9 tested letter sizes (0.25–4.0 arcmin), as a function of noise σ. Performance is pinned at 100% correct across almost 5 decades of noise — only beginning to degrade at $\sigma = 20$, roughly 730× this project's previously-calibrated value (0.027415).

This is not a sweep failure to work around: the corrected optics make the 3 mm reference aperture, at perfect focus, sharp enough that an ideal (noise-limited-only) observer reads even the smallest tested letter almost perfectly regardless of noise level, so no noise value in the pre-registered search range produces a plausible 20/20 threshold. Per this project's own discipline (documented at length in `R7acuity/RESULTS.md`), the pre-registered search bracket was *not* widened to force a number through — doing so specifically because the corrected optics broke the original bracket would itself be the kind of post-hoc tuning this project avoids. **No R7acuity sweep exists for this document, and none was fabricated.** A corrected calibration procedure is a separate, deliberate task for the future, not a quick fix folded into this sweep.

Chapter 6
Summary

Both the blur-optimization and acuity models ran cleanly, from scratch, against the corrected optics, with zero exceptions across 240 total evaluations. Both show the expected diffraction/defocus V-shape at every myopia level. They agree closely at high myopia and diverge substantially at low myopia, for a physically-understood reason (illuminance/contrast reward vs. pure blur cost). One real numerical resolution limit was found and is reported, not hidden: $r_{EE}(50\%)$ saturates at its own histogram bin width for well-focused, large-aperture conditions, where RMS radius remains the trustworthy metric. R7acuity's independent ideal-observer architecture could not be swept at all against the corrected optics — its calibration assumption is now invalidated by the same optics fix, a finding reported plainly rather than worked around. No aperture recommendation is made anywhere in this document.